



What's the Evidence for Lean?

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THE QUESTIONS OF what empirical evidence is, how it's interpreted, and how useful it is for different contexts are important to software managers and practitioners facing technology adoption decisions.¹ Independent of such evidence, however, we can't expect a technology to be universally good or universally bad—only more or less appropriate in some circumstances and for some organizations.

Given this relative appropriateness of new technologies, the claims that James Womack and his colleagues made in their influential book, *The Machine That Changed the World*,² seem quite bold. Specifically, they claimed that lean production is the one best way to manufacture cars and everything else: “lean production is a superior way for humans to make

cause lean is a software development trend that's attracted so much attention lately, we want to take a closer look at the original evidence for it. We won't make any claims directly related to lean software development or go into the literature that criticizes the appropriateness or social consequences of lean production. We only glimpse behind the curtain at the evidence underpinning the original concept of lean production and its popular interpretation, as an example of the inherent challenges of measuring and interpreting evidence for performance differences.

Interpreting Results

Basically, the evidence narrows down to interpreting results from the largest international productivity survey in the car industry's history—namely,

productivity based on the general ratio between inputs and outputs (I/O)—in this case, between labor hours and number of cars built.

Due in a large part to *The Machine That Changed the World*, the bottom-line measures of the IMVP survey have received a lot of public attention—for example, the claimed 2:1 ratio between the best Japanese productivity and worst American performers and the claimed conclusive demonstration that lean production reduced the labor hours to build cars at any level of factory automation.

However, the technicalities of calculations and interpretations leading up to these measures haven't received the same attention. Our discussion of them here is based on Dan Coffey's excellent analysis in *The Myth of Japanese Efficiency*.⁴ We also differentiate between Krafcik's survey itself³ and Womack's popular account and interpretation of it in *The Machine That Changed the World*.² Like Coffey, our main points are that the survey is worth taking seriously but that it's open to quite different interpretations from those that are generally accepted.

The evidence underpinning the original concept of lean production is MIT's IMVP survey.

things,” and “it is in everyone's interest to introduce lean production everywhere as soon as possible.”

So the question is, what's the evidence to back up these claims? Be-

MIT's International Motor Vehicle Program (IMVP) survey, carried out in the late 1980s to compare productivity across car assembly plants.³ The IMVP survey studied car assemblers'

The IMVP Survey Design

The IMVP survey deals only with measurements taken in the later assembly-plant stages of car production, which typically involve welding, painting, engine assembly, trim, and final assembly.

Womack and his colleagues justified this selection by highlighting that the IMVP survey goal was to compare assembly plants and that assembly plants are highly comparable. As Coffey pointed out, given the claims of a lean revolution, this assumption is rather astonishing because, from the outset, it

bor hours used to build a car. The first major issue that Coffey highlighted is an important yet overlooked point of the IMVP survey—that no allowance was made for work carried out in excess of a single standard shift. The survey ignored all overtime and included only a simple headcount of workers.

The IMVP survey's bottom-line productivity measure of labor hours to build a car ignored all overtime.

discounts any radical variances in plant layout or organization as a source of differences in productivity:²

[A]ssembly plants all over the world do almost exactly the same things, because practically all of today's cars and light trucks are built with very similar fabrication techniques.

Several commentators have criticized the IMVP survey's productivity measurement on the basis of how it constructs a corrected bottom line through cumulative adjustments using a number of complex indices—for example, direct and indirect workers, standard activities, standard working times, and average cars (see Karel Williams and his colleagues).⁵ However, it's easy to get lost in the complexities of these criticisms and the IMVP methodology itself and so to lose sight of the larger picture. We therefore look at only two major issues highlighted by Coffey: the way in which labor hours are measured, and the estimated levels of assembly plant automation.

Measuring Labor Hours

The IMVP survey's bottom-line productivity measure is the number of la-

bor hours used to build a car. The first major issue that Coffey highlighted is an important yet overlooked point of the IMVP survey—that no allowance was made for work carried out in excess of a single standard shift. The survey ignored all overtime and included only a simple headcount of workers.

Nevertheless, the estimated hours of labor on the single shift was divided by a daily output of cars. To see why this is a major issue when comparing I/O relationships across manufacturing sites in the automobile industry, let's take a closer look at Coffey's example.

Let's assume a plant that has 1,500 workers available to perform a single shift of 8 hours with no overtime. Suppose the daily output is 400 cars assembled. The basic productivity measure in this example would then be $(8 \times 1,500)/400 = 30$ hours per car.

Let's change the assumptions and look at a similar plant with a different configuration of shift work and overtime. Assume that overtime is allowed and that all workers currently perform a double shift. Suppose for simplicity that the headcount of available workers at this plant is 750 with the same daily output of 400 cars assembled. An accurate assessment of plant productivity in this example would then be the same as in the previous example: $(16 \times 750)/400 = 30$ hours per car. But if no allowance is made for double-shift working, and we assume that all workers in the headcount work only a single shift, this halves the time to give $(8 \times 750)/400 = 15$ hours per car.

Any survey that estimates the labor input to production by multiplying an employment total by the hours worked in a single nonovertime shift is in great danger of introducing systematic bias. The car industry is characterized by both complex shift patterns and the use of overtime. This casts serious doubts on the interpretations of the IMVP survey findings presented in *The Machine That Changed the World*—especially because the available data shows, for example, more than 10 hours' difference in the average working week in Japan versus Europe. At the time of the IMVP survey, Japanese manufacturers were known for “massive amounts of overtime.”⁴

Estimating Automation Levels

Coffey's second major issue was the interpretation of the relationship between assembly plant hours used to build a car and estimated levels of assembly plant automation. The IMVP survey data indicated that assembly plants in Japan were highly automated. Even the least-automated Japanese assembly plants surveyed still achieved a very high score on automation compared to most other car assembly plants in the world.

All else being equal, this clearly suggests that automation played a very important role in Japan's typically lower number of labor hours per car in the IMVP survey. Indeed, Womack and his colleagues found that variation in plant automation levels seemed to account for about one-third of the total variation in hours used to build a car, as measured by linear regression. However, they downplayed automation as a major determining factor, and the IMVP survey findings were accordingly interpreted in terms of a simple comparison between average hours used to assemble a car.

According to the IMVP survey, car assembly plants in Europe with high levels of automation seemed to perform

differently from plants with similar levels of automation in Japan and the rest of the world. Coffey used this fact to show how quite different inferences could be drawn from the same dataset: "Indeed, if the data for Europe had been taken separately, 'evidence' of a lean revolution would have been much harder to find,"⁴ he wrote, because Japanese plants would have been distinguished from plants elsewhere first and foremost on the basis of automation.

After making allowance for the possible distorting effects of Europe's performance, Coffey's reanalysis showed that automation would account for about three-quarters of the sampled variation. By most standards, this would be considered a very large effect size, which is why Coffey concluded that, for plants outside Europe, evidence of organizational superiority (or lean production), as opposed to high automation, would have been very hard to come by.

Assessment and Interpretation

Two different issues emerge in reinterpreting the IMVP survey findings. Coffey showed us that a reasonable case can be made for seeing Japanese car assembly plants as globally exceptional in terms of site automation but less so in terms of labor input that's not ascribable to automation. More importantly, the detail Krafcik provided on the design of the survey points to the possibility of a systematic bias in the labor input measure, one that Coffey pointed out might just as reasonably have been expected to influence both by-plant and by-region results.

Taking the collated evidence seriously, and giving due credit to the survey architects, Coffey reached a quite contrary set of conclusions, stating that⁴

First we dispute that evidence was ever presented to indicate that Japanese car assemblers possessed a discernible organizational advantage

over Western competitors once due allowance was made for differences in measured levels of assembly plant automation. Second, we show that the data in question was biased ... on this basis, we suggest a prima facie case for considering whether correcting for this bias would in fact suggest that Japanese assemblers were at this point in time struggling to convert worker effort into finished cars, vis-à-vis Western competitors.

On the basis of the IMVP survey findings, Coffey concluded that Womack might have given a flawed account of an assembler's ability to convert hours of effort into finished goods and that the survey results are better explained by a bias in the productivity measurement and the site automation level.

So, what can we learn from this? We hope we've shown that evidence isn't always what it seems to be and that popular interpretations aren't necessarily the only ones (or even the most accurate). We must go behind the bottom-line measures to reveal study assumptions, standardized models, and I/O use and adjustments. Only then can we make sense of the evidence behind a claim and its wider applicability. 🌀

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